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MILITARY SPECIFICATION

AFTERBODY TEST SET

Mk 486 MOD 0

(For Torpedo Mk 46 Mod 0/1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification covers requirements for the procurement of one type of afterbody test set, herein referred to as the test set. (See 6.1 for description and intended use.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Fiber

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-16400	Electronic Equipment, Naval Ship and Shore, General Specification

FSC 1355

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STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibration of Shipboard Equipment

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

DL 2541200	Afterbody Test Set, and all documents listed thereon
Dwg 2541198	Board, Extension
Dwg 2541204	Cable W2, Remote Control
Dwg 2541205	Cable W3, Test
Dwg 2541206	Cable W4, Auxiliary
Dwg 2541215	Gage, Backlash Measuring
Dwg 2541265	Assembly, Torque Gage

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

WD 28956	Special Manufacturing Procedure for Calibration of Afterbody Test Set

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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PUBLICATIONS

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Applications for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample shall:

(a) Be manufactured using procedures proposed for the production lot.

(b) Consist of one afterbody test set.

(c) Be tested as specified in Section 4 herein for the purpose of determining, prior to starting production, that the supplier's production methods are capable of yielding items that comply with technical requirements of the contract.

3.2 Basic requirements

3.2.1 Contractor-furnished data. No data is required by this specification, or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in manufacture of the afterbody test set shall be in accordance with requirements specified herein and with the applicable documents listed in NOSC DL 2541200.

3.3 Performance requirements and product characteristics

3.3.1 Input power. The test set shall be capable of meeting the performance requirements specified herein when 103.5 to 126.5 (115 plus or minus 11.5) volts alternating current (vac), at 57 to 63 (60 plus or minus 3) cycles per second (cps), is supplied to connector J1-A. Connector J1-C shall be the return and J1-B the ground connection.

3.3.2 Power circuits

3.3.2.1 Internal 40 volts direct current (vdc) power.

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3.3.2.1.1 Regulation. The power supply shall provide plus 39.6 to plus 40.4 vdc when provided ac input power as specified in 3.3.1 and when supplying test set internal circuits and an external load of from zero to 4.0 amperes. The OVERLOAD breaker shall trip when an external 4.0 ampere load is connected.

3.3.2.1.2 Ripple. The output ripple shall not exceed 30 millivolts rms with any combination of input voltage and load current specified in 3.3.1 and 3.3.2.1.1.

3.3.2.1.3 Adjustment. Output voltage of the power supply shall be adjustable from plus 39.98 to plus 40.02 vdc.

3.3.2.1.4 Overload protection. The power supply shall not be damaged by application of a short circuit or overload to the output.

3.3.3 Afterbody power provisions

3.3.3.1 Actuator voltage

3.3.3.1.1 An output of 37.5 to 38.5 vdc shall be supplied with the SPEED switch in HI position.

3.3.3.1.2 An output of 11.5 to 14.5 vdc shall be supplied with the SPEED switch in LO position.

3.3.4 Test functions

3.3.4.1 Self test provisions. SELF TEST switch indications shall be in accordance with Table I.

Table I

SELF TEST Functions

SELF TEST Switch Position	Function	Indications
OFF	Interlock with TEST SELECT for normal operation	None
1 GO	Power supply output between 39.0 and 41.0 vdc	GO lamp illuminates
2 GO	Simulates 1 ohm resistance to continuity evaluator	GO lamp illuminates
3 GO	Simulated 1.3 megohms resistance to leakage evaluator	GO lamp illuminates

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Table I (Cont)

SELF TEST Functions

SELF TEST Switch Position	Function	Indications
4 NO GO	Simulates 7 ohm resistance to continuity evaluator	NO GO lamp illuminates
5 NO GO	Simulates 0.91 megohm resistance to leakage evaluator	NO GO lamp illuminates
6 NO GO	Simulates zero actuator current to actuator current and timing evaluators. Requires SPEED at HI, SELECT at OFF, Operate DIRECTION to LEFT/UP or RIGHT/DOWN	NO GO TEST IN PROGRSSS, and appropriate DIRECTION lamps illuminate for an interval of from two to four seconds. TEST IN PROGRESS lamp extinguishes, NO GO and appropriate DIRECTION lamp remain illuminated.
7 NO GO	Simulates out-of limits actuator current to actuator current and timing evaluator. Requires SPEED at HI, SELECT at OFF. Operate DIRECTION to LEFT/UP or RIGHT/DOWN	GO lamp illuminates for an instant. NO GO, TEST IN PROGRESS, appropriate DIRECTION lamps illuminate for an interval of from two to four seconds. TEST IN PROGRESS lamp extinguishes, NO GO and appropriate DIRECTION lamp remain illuminated.

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Table I (Cont)

SELF TEST Functions

SELF TEST Switch Position	Function	Indications
8 SEE INST	Simulates within limits actuator current to actuator current and timing evaluator. Requires SPEED at HI, SELECT at OFF. Operate DIRECTION to LEFT/UP or RIGHT/DOWN	GO, TEST IN PROGRESS and appropriate DIRECTION lamps illuminate for an interval of from two to four seconds
	Current is interrupted at completion of test timing cycle which simulates out of limits timing test to time evaluator	TEST IN PROGRESS and GO lamps extinguish, appropriate DIRECTION lamp remains illuminated NO-GO lamp illuminates.

3.3.4.2 CONTINUITY-1 test provisions. The resistance and continuity tests provided in positions 1 through 23 of the CIRCUIT SELECT switch with CONTINUITY-1 position selected on TEST SELECT shall be as indicated in Table II.

Table II

CONTINUITY-1 Test Functions

CIRCUIT SELECT Switch Position	Function	NO GO below	Resistance Values (Ohms) vs Lamp Indications		
			Change to GO	GO	Change to NO GO
1	Upper Rudder	6.3	6.3-7.7	7.7-38	38-42
2	Lower Rudder	6.3	6.3-7.7	7.7-38	38-42
3	Elevator	6.3	6.3-7.7	7.7-66.5	66.5-73.5
4	Alternator +40 vdc	18.8	18.8-21.2	21.2-43.0	43.0-47.0

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Table II (Cont)

CONTINUITY-1 Test Functions

CIRCUIT SELECT Switch Position	Function	NO GO below	Resistance Values (Ohms) vs Lamp Indications		
			Change to GO	GO	Change to NO GO
5	Alternator -40 vdc	18.8	18.8-21.2	21.2-43.0	43.0-47.0
6	Alternator AC	9.0	9.0-11.0	11.0-18.5	18.5-21.5
7	Alternator Field	8.0	8.0-9.0	9.0-11.0	11.0-12.0
8	Alternator Jumper	N/A*	N/A	0.0-1.0	1.0 to 5.0
9	Alternator Jumper	N/A	N/A	0.0-1.0	1.0 to 5.0
10	Alternator Jumper	N/A	N/A	0.0-1.0	1.0 to 5.0
11	Alternator Jumper	N/A	N/A	0.0-1.0	1.0 to 5.0
12	Alternator Jumper	N/A	N/A	0.0-1.0	1.0 to 5.0
13	Alternator Jumper	N/A	N/A	0.0-1.0	1.0 to 5.0
14	Alternator Jumper	N/A	N/A	0.0-1.0	1.0 to 5.0
15	Alternator Continuity	N/A	N/A	0.0-1.0	1.0 to 5.0
16	Warm Plug Gyro Unit	N/A	N/A	0.0-1.0	1.0 to 5.0
17	Igniter	N/A	N/A	0.0-1.0	1.0 to 5.0
18	Seawater Battery	N/A	N/A	0.0-1.0	1.0 to 5.0
19	Actuator Continuity	N/A	N/A	0.0-1.0	1.0 to 5.0
20	Spare	N/A	N/A	0.0-1.0	1.0 to 5.0
21	Upper Rudder	N/A	N/A	0.0-1.0	1.0 to 5.0
22	Lower Rudder	N/A	N/A	0.0-1.0	1.0 to 5.0

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Table II (Cont)

CONTINUITY-1 Test Functions

CIRCUIT SELECT Switch Position	Function	NO GO below	Resistance Values (Ohms) vs Lamp Indications		
			Change to GO	GO	Change to NO GO
23	Elevator	N/A	N/A	0.0-1.0	1.0 to 5.0
24	OFF				

3.3.4.3 CONTINUITY-2 test provisions. The continuity tests provided in positions 1 through 23 of the CIRCUIT SELECT switch with CONTINUITY-2 position selected on TEST SELECT switch shall be as indicated in Table III.

Table III

CONTINUITY-2 Test Functions

CIRCUIT SELECT Switch Position	Function	Resistance Values (Ohms) vs Lamp Indications	
		GO	Change to NO GO
1	Warm Plug Air/ASROC	0.0-1.0	1.0-5.0
2	Fire Control Common	0.0-1.0	1.0-5.0
3	Ceiling Sensor	0.0-1.0	1.0-5.0
4	Depth Sensor	0.0-1.0	1.0-5.0
5	Depth Drive	0.0-1.0	1.0-5.0
6	Ceiling Drive	0.0-1.0	1.0-5.0
7	Tube Launch	0.0-1.0	1.0-5.0
8	Common & Continuity	0.0-1.0	1.0-5.0
9	Jumper	0.0-1.0	1.0-5.0
10	Continuity	0.0-1.0	1.0-5.0
11	Instrument Wiring	No Test	---

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Table III

CONTINUITY-2 Test Functions

CIRCUIT SELECT Switch Position	Function	Resistance Values (Ohms) vs Lamp Indications	
		GO	Change to NO GO
12	Instrument Wiring	No Test	---
13	Instrument Wiring	No Test	---
14	Instrument Wiring	No Test	---
15	Instrument Wiring	No Test	---
16	Igniter	0.0-1.0	1.0-5.0
17	Igniter	0.0-1.0	1.0-5.0
18	Seawater Battery	0.0-1.0	1.0-5.0
19	Spare	No Test	---
20	Spare	No Test	---
21	Spare	No Test	---
22	Spare	No Test	---
23	Spare	No Test	---
24	OFF		

3.3.4.4 Leakage test provisions. Leakage resistance tests of afterbody circuits to ground shall be performed internally through functions of TEST SELECT switch in positions LEAKAGE-1 and LEAKAGE-2. Positions 1 through 23 of CIRCUIT SELECT switch shall select afterbody circuits for test in each of LEAKAGE-1 and LEAKAGE-2 positions.

3.3.4.4.1 LEAKAGE-1 test provisions. Leakage resistance tests provided in positions 1 through 23 of CIRCUIT SELECT switch with LEAKAGE-1 position selected on TEST SELECT switch shall be as indicated in Table IV.

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Table IV

LEAKAGE-1 Test Functions

CIRCUIT SELECT Switch Position	Function	Resistance Values (Megohms) vs Lamp Indications NO GO lamp illuminates at 0.95 megohm and changes to GO indication before 1.10 megohms is reached
1	Upper Rudder	From 0.96 to 1.10
2	Lower Rudder	From 0.96 to 1.10
3	Elevator	From 0.96 to 1.10
4	Alternator +40 vdc	From 0.96 to 1.10
5	Alternator -40 vdc	From 0.96 to 1.10
6	Alternator ac	From 0.96 to 1.10
7	Alternator Field	From 0.96 to 1.10
8	Alternator Jumper	From 0.96 to 1.10
9	Alternator Jumper	From 0.96 to 1.10
10	Alternator Jumper	From 0.96 to 1.10
11	Alternator Jumper	From 0.96 to 1.10
12	Alternator Jumper	From 0.96 to 1.10
13	Alternator Jumper	From 0.96 to 1.10
14	Alternator Jumper	From 0.96 to 1.10
15	Alternator Continuity	From 0.96 to 1.10
16	Warm Plug Gyro Uncage	From 0.96 to 1.10
17	Igniter	From 0.96 to 1.10
18	Seawater Battery	From 0.96 to 1.10
19	Actuator Continuity	From 0.96 to 1.10

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Table IV (Cont)

LEAKAGE-1 Test Functions

CIRCUIT SELECT Switch Position	Function	Resistance Values (Megohms) vs Lamp Indications NO GO lamp illuminates at 0.95 megohm and changes to GO indication before 1.10 megohms is reached
20	Spare	N/A
21	Upper Rudder	From 0.96 to 1.10
22	Lower Rudder	From 0.96 to 1.10
23	Elevator	From 0.96 to 1.10
24	OFF	

3.3.4.4.2 LEAKAGE-2 test provisions. Leakage resistance tests provided in positions 1 through 23 of CIRCUIT SELECT switch with LEAKAGE-2 selected on TEST SELECT switch shall be as indicated in Table V.

Table V

LEAKAGE-2 Test Functions

CIRCUIT SELECT Switch Position	Function	Resistance Values (Megohms) vs Lamp Indications NO GO lamp illuminates at 0.95 megohm and changes to GO indication before 1.10 megohms is reached
1	Warm Plug Air/ASROC	From 0.96 to 1.10
2	Fire Control Comm	From 0.96 to 1.10
3	Ceiling Sensor	From 0.96 to 1.10
4	Depth Sensor	From 0.96 to 1.10
5	Depth Drive	From 0.96 to 1.10
6	Ceiling Drive	From 0.96 to 1.10

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Table V (Cont)

LEAKAGE-2 Test Functions

CIRCUIT SELECT Switch Position	Function	Resistance Values (Megohms) vs Lamp Indications NO GO lamp illuminates at 0.95 megohm and changes to GO indication before 1.10 megohms is reached
7	Tube Launch	From 0.96 to 1.10
8	Common & Continuity	From 0.96 to 1.10
9	Jumper	From 0.96 to 1.10
10	Continuity	From 0.96 to 1.10
11	Instrument Wiring	From 0.96 to 1.10
12	Instrument Wiring	From 0.96 to 1.10
13	Instrument Wiring	From 0.96 to 1.10
14	Instrument Wiring	From 0.96 to 1.10
15	Instrument Wiring	From 0.96 to 1.10
16	Igniter	From 0.96 to 1.10
17	Igniter	From 0.96 to 1.10
18	Seawater Battery	From 0.96 to 1.10
19	Spare	N/A
20	Spare	N/A
21	Spare	N/A
22	Spare	N/A
23	Spare	N/A
24	OFF	

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3.3.4.5 ACTUATOR CONTROL test provisions

3.3.4.5.1 SPEED switch. The HI position shall provide 38.0 plus or minus 0.5 vdc output; the LO position shall provide 13.0 plus or minus 1.5 vdc output.

3.3.4.5.2 DIRECTION switch. It shall be positioned to LEFT/UP or RIGHT/DOWN to facilitate performance evaluation of the actuator under test. The LEFT/UP or RIGHT/DOWN lamps shall illuminate in the corresponding positions of DIRECTION switch. The remote control shall be energized when the DIRECTION switch is in OFF REMOTE and the SPEED switch is in LO position.

3.3.4.5.3 Power control functions. The SELECT and DIRECTION controls shall provide power to the actuator motors and lamp indications as specified in Table VI.

Table VI

ACTUATOR CONTROLS Test Functions

ACTUATOR Switch Position		Lamp Indication	Afterbody Connections Actuator Voltage	
DIRECTION	SELECT		Positive	Negative
1 LEFT/UP	UPPER RUDDER	LEFT/UP illuminates	P5520-C	P5520-B
2 RIGHT/DOWN	UPPER RUDDER	RIGHT/DOWN illuminates	P5520-B	P5520-A
3 RIGHT/DOWN	LOWER RUDDER	RIGHT/DOWN illuminates	P5520-L	P5520-K
4 LEFT/UP	LOWER RUDDER	LEFT/UP illuminates	P5520-K	P5520-J
5 LEFT/UP	ELEVATOR	LEFT/UP illuminates	P5520-F	P5520-E
6 RIGHT/DOWN	ELEVATOR	RIGHT/DOWN illuminates	P5520-G	P5520-F

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3.3.4.5.4 A remote control feature to provide a facility for positioning afterbody control surfaces while being viewed at close range shall be a feature of the test set and available through connection to J2. Two pushbutton switches mounted on a hand size enclosure at the end of a suitable cable shall duplicate the functions of the DIRECTION switch with low voltage (3.3.3.1.2) applied. Remote control switch functions are detailed in Table VII.

Table VII

Remote Control Unit Actuator Direction Function

Actuator Selected	LEFT/UP	RIGHT/DOWN
Upper Rudder	LEFT	RIGHT
Lower Rudder	LEFT	RIGHT
Elevator	UP	DOWN

3.3.4.6 DC current source. A direct current cource, to be provided from the test set internal 40 vdc power supply, shall be controlled by the FIELD FLASH pushbutton switch. It shall be applied to the afterbody at connector P5510 pins R (positive) and N (return). Source resistance shall be 45.3 ohms plus or minus 3 percent. This circuit shall be capable of delivering between 850 to 910 milliamperes to a zero ohm load, or 700 to 750 milliamperes to a 10 ohm load.

3.3.4.7 Actuator performance test and evaluation. Internal circuits shall sense the running current amplitude and time required for the actuator to operate between limits switch control. The evaluator shall control the indications provided by the GO, TEST IN PROGRESS, and NO GO indicators.

3.3.4.7.1 Test indication. The TEST IN PROGRESS lamp shall illuminate during the period power is applied to the actuator under test. To prevent misreading GO-NO GO lamps, the TEST IN PROGRESS lamp shall illuminate when the SPEED switch is in LO position.

3.3.4.7.2 Actuator current evaluation. The evaluator shall produce a NO GO indication when the average running current of the actuator under test exceeds the limit specified in Table VIII. The starting current pulse shall not be evaluated for a period of 60 to 100 milliseconds after application of actuator power.

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Table VIII

Actuator Performance Limits

Actuator Select	CURRENT LIMITS	TIME LIMITS (Milliseconds)	
		LOW	HI
UPPER RUDDER	930 to 980 ma	690 to 710	1035 to 1065
LOWER RUDDER	930 to 980 ma	690 to 710	1035 to 1065
ELEVATORS	240 to 260 ma	690 to 710	1035 to 1065

3.3.4.7.3 Actuator running time evaluation. The evaluator shall produce a NO GO indication when the actuator dc current source is interrupted either before or after the time period defined by the limits specified in Table VIII.

3.3.4.7.4 Indication display time. Either the GO or NO GO lamp shall remain lit until operating controls are readjusted.

3.3.4.8 Indicator lamp test. All indicator lamps, except TEST SET POWER, used in the test set shall have a press-to-test feature incorporated to allow the operator to determine that each indicator lamp is operating.

3.3.5 Auxiliary equipment. The following auxiliary equipment shall be supplied with the test set:

(a) Backlash Measuring Gage 2541215 shall be used for the measurement of backlash encountered during tests utilizing this test set.

(b) Torque Gage 2541265 shall be used to apply calibrated torquing force to trailing edge of control surface for measurement of backlash and splay.

(c) Extension Board 2541198 shall be used to extend plug-in modules above enclosures to facilitate troubleshooting of circuit.

(d) Remote Control Cable W2 2541204 for remote LEFT/UP and DOWN/RIGHT operation.

(e) Auxiliary Cable W4 2541206, when required, shall be connected to Cable W3 2541205 for special tests.

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3.3.6 Service life. The service life of the test set shall be 5 years of service with a minimum of 60 days between calibrations.

3.4 Environmental requirements. Unless otherwise specified the test set shall be capable of conforming to the performance and product characteristics of this specification after being exposed, at any temperature in the applicable range, to the environmental requirements under all conditions specified in this section.

3.4.1 Vibration. The test set shall be capable of withstanding the Type 1 vibration required by MIL-STD-167, except that the variable frequency and endurance test requirements shall be as follows: The test set in operating condition shall be vibrated from 5 to 15 cps at a half peak amplitude of 0.030 plus or minus 0.006 inch and from 16 to 25 cps at a half peak amplitude of 0.020 plus or minus 0.004 inch. The vibration shall be applied in discrete intervals of one cps and applied for one minute at each interval.

3.4.2 Shock. The test set shall be capable of withstanding the shock required by MIL-S-901, for Grade B, Class I, lightweight equipment, Type A, in the nonoperating condition, except that the height of the hammer drops shall be: 1 foot, 1-1/2 feet, and 3 feet.

3.4.3 Inclination. The test set shall be capable of meeting the inclination requirements specified in MIL-E-16400.

3.4.4 Temperature. The test set shall be capable of withstanding the temperatures specified in MIL-E-16400 for Class 4 equipments.

3.4.5 Humidity. The test set shall be capable of withstanding the humidity requirements specified in MIL-E-16400.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation consistent with the requirements of this specification. All parts used in the test set shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as

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otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections of the test set shall be classified as follows:

(a) Preproduction inspection

(1) Preproduction sample (see 4.3.1)

(2) Preproduction tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

(c) Acceptance inspection

(1) Acceptance sampling (see 4.3.3)

(2) Acceptance tests and examinations (see 4.7)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample manufactured in accordance with 3.1, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the test set by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production samples shall be selected from units having satisfactorily passed acceptance tests and examinations of 4.7. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

(a) Step 1. Upon initiation of production select one unit at random from each group of 20 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2. Each sample so selected shall be numbered consecutively, starting with Sample Number 1.

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(b) Step 2. Select one unit at random from each group of 50 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively, starting with a sample number one greater than the highest sample number selected in Step 1.

4.3.2.1 Defective units. When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval the proposed corrective action to prevent recurrence of the defect.

(d) Submit to the procuring activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

(e) Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Acceptance sampling. The tests and examinations specified in 4.7 shall be conducted on each test set to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the test set shall be subjected to tests under the following conditions:

(a) Temperature	Room ambient (60 to 95°F)
(b) Humidity (Relative)	Room ambient, up to 95%
(c) Altitude	Normal ground atmospheric pressure

4.5 Preproduction tests. The test set submitted as a preproduction sample shall be subjected to all preproduction tests specified in 4.5.2. Prior to and immediately after the preproduction tests the test set shall be subjected to the acceptance tests and examinations (see 4.7). Certified inspection data covering the results of preproduction tests shall be forwarded to the Naval Ordnance Systems Command. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests.

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(b) Run or test plan used.

(c) Serial number of the test set.

(d) Test results of preproduction tests.

4.5.1 Preproduction test equipment. The following items of test equipment are required to perform the preproduction tests specified herein:

4.5.1.1 A vibration test fixture capable of applying vibration as specified in 3.4.1.

4.5.1.2 A shock fixture capable of applying shocks specified in 3.4.2.

4.5.1.3 An inclination testing device capable of performing the inclination tests specified in MIL-E-16400.

4.5.1.4 A humidity-temperature chamber capable of applying the humidity and temperature tests specified in MIL-E-16400.

4.5.1.5 Items of test equipment specified in 4.7.2.

4.5.2 Preproduction tests

4.5.2.1 Vibrate the test set in accordance with vibration schedules required by MIL-STD-167 with deviations specified in 3.4.1.

4.5.2.2 Shock. Secure the test set to the shock test fixture and apply shocks as specified in MIL-S-901 with deviations specified in 3.4.2.

4.5.2.3 Inclination. Subject the test set to the inclination tests specified in MIL-E-16400.

4.5.2.4 Temperature. Subject the test set to temperature tests, both operating and nonoperating, as specified in MIL-E-16400 for Class 4 equipments.

4.5.2.5 Humidity. Subject the test set to the humidity tests specified in MIL-E-16400.

4.6 Periodic production tests

4.6.1 Periodic production test equipment. The following items of test equipment, or equivalent items, will be required to perform the tests specified herein:

(a) Temperature chamber capable of providing the environment specified in 3.4.4.

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(b) Vibration test fixture capable of applying vibration as specified in 3.4.1.

4.6.2 Periodic production tests. Each periodic production sample, after satisfactorily passing the acceptance tests and examination (4.7) shall be subjected to environmental tests for temperature (4.5.2.4) and for vibration (4.5.2.1). Unless otherwise specified, the sample shall be classed as defective if it fails to pass satisfactorily the acceptance tests and examinations subsequent to these environmental temperature and vibration tests.

4.7 Acceptance tests and examinations. Unless otherwise specified, each test set delivered under contract shall be subjected to the tests and examinations specified herein.

4.7.1 Conditions

4.7.1.1 Test sequence. Unless otherwise noted, the sequence of testing procedures presented herein is mandatory.

4.7.1.2 Test results. Where space is provided in the Appendix Report Form to indicate value or specific indication obtained, the specific indication shall be recorded if a limit or limits are given in the Limits column for the test. If limits are not stated in the Limits column, the individual performing the test or inspection shall initial in spaces provided in the Reading column, indicating that requirements were satisfactorily met.

4.7.1.3 Acceptance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units which have not been corrected will be considered for acceptance only upon formal application by the contractor to the Naval Ordnance Systems Command via the cognizant Government representative.

4.7.2 Acceptance test equipment. Items of test equipment listed in Table IX are required to perform acceptance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals. Procedures for and results of all calibrations shall be subject to approval of the cognizant Government inspector. Items having equivalent characteristics may be substituted where necessary. Verification shall be made that all test equipment used is identified as being within calibration.

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Table IX

Acceptance Test Equipment

Item	Identification
Digital dc voltmeter (Range: 0.000 to 9.999 vdc; 00.00 to 99.99 vdc; input impedance: 11 megohms; accuracy: 0.2% of applied voltage or ±2 digits whichever is greater; sampling rate: maximum 4/seconds to minimum 1/second, variable.)	Hewlett-Packard Model 405B
Oscilloscope (Bandpass: dc to 15 megacycles; Rise time: 0.023 micro-seconds; accuracy: 3% from 0.05 to 20 volts/cm.)	Tektronix Model 535A with CA preamplifier
Electronic counter (Range: 10 cps to 120 kilocycles; Accuracy: ±1 count.)	Hewlett-Packard Model 522B
DC vacuum tube volt-ohm-ammeter voltmeter (Range: 1 mv to 1000 v full scale at accuracy of ±1% of full scale; Ammeter: Range: 1 μamp to 1 amp; ±2% of full scale; input resistance from 1000 ohms at 1 μamp to 0.1 ohm at 1 amp.)	Hewlett-Packard Model 412A
Resistance decade box (Low resistance 0.1 ohm to 110 ohms at 0.1 to 1.0 ohm ±0.5%, 1.0 to 10.0 ohm ±0.15%, and 10.0 to 110 ohm ±0.05%.)	General Radio Model 1432K
Resistance decade box (High resist-ance 1000 ohms to 11 megohms at 0.05% tolerance.)	General Radio Model 1432Y
Resistors, loading (All ±0.25%)	Commercial
38.5 ohms, 20 watts	
41.0 ohms, 20 watts	
146.0 ohms, 10 watts	
152.0 ohms, 10 watts	

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4.7.3 Electrical tests

4.7.3.1 Initial test setup

4.7.3.1.1 Control settings. Position test set panel controls to the following initial settings:

Control	Position
CIRCUIT SELECT	OFF
TEST SELECT	SELF TEST
SELF TEST	OFF
SELECT	OFF
DIRECTION	OFF REMOTE
SPEED	HI
POWER	OFF
ON OVERLOAD	ON

4.7.3.2 Equipment turn-on

4.7.3.2.1 Connect P1 POWER cable to J1 POWER connector on front panel and connect plug at other end of cable to a 60 cycle 110 to 120 (115 ±5.0) vac power source.

4.7.3.2.2 Connect P3 TEST cable to J3 TEST connector on front panel. Connectors at the other end of cable are to remain unconnected (open).

4.7.3.2.3 Connect P2 REMOTE cable with remote control at other end, to J2 REMOTE connector on the front panel.

4.7.3.2.4 Set POWER switch in test set front panel to ON. POWER lamp (red) shall illuminate. Allow test set to warm up for two minutes.

4.7.3.3 Self test

4.7.3.3.1 Position SELF TEST switch to GO 1. Connect + (plus) voltage lead of digital dc voltmeter to +40V test point, and connect common lead to COMMON test point.

4.7.3.3.2 GO lamp (green) shall illuminate. Digital dc voltmeter shall indicate +39.6 to +40.4 vdc.

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4.7.3.3.3 Disconnect digital dc voltmeter.

4.7.3.3.4 Position SELF TEST switch to GO 2.

4.7.3.3.5 GO lamp shall light, indicating less than one ohm resistance at continuity evaluator.

4.7.3.3.6 Set SELF TEST switch to GO 3.

4.7.3.3.7 GO lamp shall remain lit, indicating more than 1.10 megohms resistance at leakage evaluator.

4.7.3.3.8 Position SELF TEST switch to NO GO 4.

4.7.3.3.9 NO GO lamp (red) shall illuminate, indicating more than 5 ohms resistance at continuity evaluator.

4.7.3.3.10 Set SELF TEST switch to NO GO 5.

4.7.3.3.11 NO GO lamp shall remain illuminated, indicating less than 0.95 megohm resistance at leakage evaluator.

4.7.3.3.12 Position SELF TEST switch to NO GO 6.

4.7.3.3.13 Connect dc vtvm to ACTUATOR test point and to COMMON test point.

4.7.3.3.14 Position DIRECTION switch to LEFT/UP.

4.7.3.3.15 NO GO, TEST IN PROGRESS (amber), and LEFT/UP (amber) lamps shall illuminate. After an elapsed time of from two to four seconds TEST IN PROGRESS shall extinguish, NO GO lamp and LEFT/UP lamp shall remain illuminated. DC vtvm shall indicate from +0.01 vdc to -0.01 vdc while TEST IN PROGRESS lamp is illuminated.

4.7.3.3.16 Position SELF TEST switch to NO GO 7 and DIRECTION switch to RIGHT/DOWN.

4.7.3.3.17 GO lamp shall flash momentarily. NO GO, TEST IN PROGRESS, and RIGHT/DOWN (amber) lamps shall illuminate. After an elapsed time of from two to four seconds TEST IN PROGRESS lamp shall extinguish. NO GO lamp and RIGHT/DOWN lamp shall remain illuminated. DC vtvm shall indicate +1.3 to +1.5 vdc while TEST IN PROGRESS lamp is illuminated.

4.7.3.3.18 Position SELF TEST switch to SEE INST 8 and DIRECTION switch to LEFT/UP.

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4.7.3.3.19 GO, TEST IN PROGRESS, and LEFT/UP lamps shall illuminate. After an elapsed time of from two to four seconds TEST IN PROGRESS and GO lamps shall extinguish, NO GO lamp shall illuminate and LEFT/UP shall remain lit. DC vtvm shall indicate 800 to 1000 millivolts while TEST IN PROGRESS lamp is illuminated.

4.7.3.3.20 Disconnect dc vtvm.

4.7.3.3.21 Position SELF TEST switch to OFF and DIRECTION switch to OFF REMOTE.

4.7.3.4 Continuity test

4.7.3.4.1 Position TEST SELECT switch to CONTINUITY-1.

4.7.3.4.2 Connect a resistance decade box with a range from 0.1 ohm to 110 ohms across the connector pins at each CIRCUIT SELECT switch position as indicated in Table X. Increase the resistance value in 0.1 ohm increments from left column indication noted to right column indication to ascertain that lamp indication agrees with resistance value noted at each position shown in Table X.

4.7.3.4.3 Position TEST SELECT switch to CONTINUITY-2.

4.7.3.4.4 Connect the resistance decade box used in 4.7.3.4.2 across the connector pins at each position of CIRCUIT SELECT switch indicated in Table XI. Increase resistance values in 0.1 ohm increments from 0.0 to 5.0 ohms to ascertain that lamp indication agrees with resistance value specified at each position as shown in Table XI.

4.7.3.4.5 Disconnect resistance decade box leads.

4.7.3.5 Leakage tests

4.7.3.5.1 Position TEST SELECT switch to LEAKAGE-1.

4.7.3.5.2 Connect a resistance decade box with a range exceeding 0.95 megohms to 1.10 megohms across the connector pins at each CIRCUIT SELECT switch position indicated in Table XII. Increase the resistance value in 10K ohm increments to check lamp indications at each position shown in Table XII.

4.7.3.5.3 Position TEST SELECT switch to LEAKAGE-2. Connect resistance decade box noted in 4.7.3.5.2 across connector pins at each CIRCUIT SELECT switch position indicated in Table XIII. Increase the resistance value in 10K ohm increments to check lamp indications at each position shown in Table XIII.

4.7.3.5.4 Disconnect resistance decade box from all pin connections.

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Table X

CONTINUITY-1 Test

CIRCUIT SELECT Switch Position	Function	Connector Pins		Resistance Values (Ohms) vs Lamp Indications			
		From	To	NO GO Below	Change to GO	GO	Change to NO GO
1	Upper Rudder	P5520 A	P5520 B	6.3	6.3-7.7	7.7-38.0	38.0-42.0
2	Lower Rudder	P5520 J	P5520 K	6.3	6.3-7.7	7.7-38.0	38.0-42.0
3	Elevator	P5520 E	P5520 F	6.3	6.3-7.7	7.7-66.5	66.5-73.5
4	Alternator +40 vdc	P5510 S	P5510 A	18.8	18.8-21.2	21.2-43.0	43.0-47.0
5	Alternator -40 vdc	P5510 D	P5510 S	18.8	18.8-21.2	21.2-43.0	43.0-47.0
6	Alternator ac	P5510 S	P5510 B	9.0	9.0-11.0	11.0-18.5	18.5-21.5
7	Alternator Field	P5510 R	P5510 N	8.0	8.0-9.0	9.0-11.0	11.0-12.0
8	Alternator Jumper	P5510 E	P5510 H	N/A	N/A	0-1.0	1.0-5.0
9	Alternator Jumper	P5510 E	P5510 J	N/A	N/A	0-1.0	1.0-5.0
10	Alternator Jumper	P5510 E	P5510 A	N/A	N/A	0-1.0	1.0-5.0
11	Alternator Jumper	P5510 C	P5510 B	N/A	N/A	0-1.0	1.0-5.0
12	Alternator Jumper	P5510 K	P5510 L	N/A	N/A	0-1.0	1.0-5.0

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Table X (Cont)

CONTINUITY-1 Test

CIRCUIT SELECT Switch Position	Function	Connector Pins		Resistance Values (Ohms) vs Lamp Indications			
		From	To	NO GO Below	Change to GO	GO	Change to NO GO
13	Alternator Jumper	P5510 K	P5510 M	N/A	N/A	0-1.0	1.0-5.0
14	Alternator Jumper	P5510 K	P5510 D	N/A	N/A	0-1.0	1.0-5.0
15	Alternator Continuity	P5510 F	P5510 G	N/A	N/A	0-1.0	1.0-5.0
16	Warm Plug Gyro Uncg.	P5520 h	P5608 0	N/A	N/A	0-1.0	1.0-5.0
17	Igniter	P5520 j	P5629 1	N/A	N/A	0-1.0	1.0-5.0
18	Seawater Battery	P5520 V	P5626 A	N/A	N/A	0-1.0	1.0-5.0
19	Actuator Continuity	P5520 M	P5520 N	N/A	N/A	0-1.0	1.0-5.0
20	Spare	N/C*	N/C	N/A	N/A	0-infinity	N/A
21	Upper Rudder	P5520 A	P5520 C	N/A	N/A	0-1.0	1.0-5.0
22	Lower Rudder	P5520 J	P5520 L	N/A	N/A	0-1.0	1.0-5.0
23	Elevator	P5520 E	P5520 G	N/A	N/A	0-1.0	1.0-5.0
24	OFF	OFF	OFF				

* No Connection

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Table XI

CONTINUITY-2 Test

CIRCUIT SELECT Switch Position	Function	Connector Pins		Resistance Values (Ohms) vs Lamp Indications	
		From	To	GO	NO GO
1	Warm Plug Air/ASROC	P5520 m	P5608 A	0.0-1.0	1.0-5.0
2	Fire Control Common	P5520 a	P5608 B	0.0-1.0	1.0-5.0
3	Ceiling Sensor	P5520 b	P5608 C	0.0-1.0	1.0-5.0
4	Depth Sensor	P5520 c	P5608 D	0.0-1.0	1.0-5.0
5	Depth Drive	P5520 d	P5608 E	0.0-1.0	1.0-5.0
6	Ceiling Drive	P5520 e	P5608 F	0.0-1.0	1.0-5.0
7	Tube launch	P5520 f	P5608 M	0.0-1.0	1.0-5.0
8	Common & Continuity	P5520 g	P5608 N	0.0-1.0	1.0-5.0
9	Jumper	P5520 a	P5608 N	0.0-1.0	1.0-5.0
10	Continuity	P5520 n	P5608 P	0.0-1.0	1.0-5.0
11	Instrument Wiring	P5520 P	N/C	0-infinity	N/A
12	Instrument Wiring	P5520 R	N/C	0-infinity	N/A

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Table XI (Cont)

CONTINUITY-2 Test

CIRCUIT SELECT Switch Position	Function	Connector Pins		Resistance Values (Ohms) vs Lamp Indications	
		From	To	GO	NO GO
13	Instrument Wiring	P5520 S	N/C	0-infinity	N/A
14	Instrument Wiring	P5520 T	N/C	0-infinity	N/A
15	Instrument Wiring	P5520 U	N/C	0-infinity	N/A
16	Igniter	P5520 i	P5629 2	0.0-1.0	1.0-5.0
17	Igniter	P5520 i	P5629 2	0.0-1.0	1.0-5.0
18	Seawater Battery	P5520 W	P5626 B	0.0-1.0	1.0-5.0
19	Spare	N/C	N/C	N/A	N/A
20	Spare			N/A	N/A
21	Spare			N/A	N/A
22	Spare			N/A	N/A
23	Spare	N/C	N/C	N/A	N/A
24	OFF	OFF	OFF		

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Table XII

LEAKAGE-1 Test

CIRCUIT SELECT Switch Position	Function	Afterbody Connector Pins		Resistance Value vs Lamp Indication
		From	To	NO GO lamp illuminates at 0.95 megohm and indication changes to GO lamp before 1.10 megohms is reached.
1	Upper Rudder	P5520 A	COMMON	From 0.96 to 1.10
2	Lower Rudder	P5520 J	COMMON	From 0.96 to 1.10
3	Elevator	P5520 E	COMMON	From 0.96 to 1.10
4	Alternator +40 vdc	P5510 S	COMMON	From 0.96 to 1.10
5	Alternator -40 vdc	P5510 D	COMMON	From 0.96 to 1.10
6	Alternator ac	P5510 S	COMMON	From 0.96 to 1.10
7	Alternator Field	P5510 R	COMMON	From 0.96 to 1.10
8	Alternator Jumper	P5510 E	COMMON	From 0.96 to 1.10
9	Alternator Jumper	P5510 E	COMMON	From 0.96 to 1.10
10	Alternator Jumper	P5510 E	COMMON	From 0.96 to 1.10
11	Alternator Jumper	P5510 C	COMMON	From 0.96 to 1.10
12	Alternator Jumper	P5510 K	COMMON	From 0.96 to 1.10

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Table XII (Cont)

LEAKAGE-1 Test

CIRCUIT SELECT Switch Position	Function	Afterbody Connector Pin		Resistance Value vs Lamp Indication
		From	To	NO GO lamp illuminates at 0.95 megohm and indication changes to GO lamp before 1.10 megohms is reached.
13	Alternator Jumper	P5510 K	COMMON	From 0.96 to 1.10
14	Alternator Jumper	P5510 K	COMMON	From 0.96 to 1.10
15	Alternator Continuity	P5510 F	COMMON	From 0.96 to 1.10
16	Warm Plug Gyro Uncg.	P5520 h	COMMON	From 0.96 to 1.10
17	Igniter	P5520 j	COMMON	From 0.96 to 1.10
18	Seawater Battery	P5520 V	COMMON	From 0.96 to 1.10
19	Actuator Continuity	P5520 M	COMMON	From 0.96 to 1.10
20	Spare	N/C	N/C	0-infinity
21	Upper Rudder	P5520 A	COMMON	From 0.96 to 1.10
22	Lower Rudder	P5520 J	COMMON	From 0.96 to 1.10
23	Elevator	P5520 E	COMMON	From 0.96 to 1.10
24	OFF	OFF	OFF	

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Table XIII

LEAKAGE-2 Test

CIRCUIT SELECT Switch Position	Function	Afterbody Connector Pin		Resistance Value vs Lamp Indication
		From	To	NO GO lamp illuminates at 0.95 megohm and indication changes to GO lamp before 1.10 megohms is reached.
1	Warm Plug Air/ASROC	P5520 m	COMMON	From 0.96 to 1.10
2	Fire Control Comm.	P5520 a	COMMON	From 0.96 to 1.10
3	Ceiling Sensor	P5520 b	COMMON	From 0.96 to 1.10
4	Depth Sensor	P5520 c	COMMON	From 0.96 to 1.10
5	Depth Drive	P5520 d	COMMON	From 0.96 to 1.10
6	Ceiling Drive	P5520 e	COMMON	From 0.96 to 1.10
7	Tube Launch	P5520 f	COMMON	From 0.96 to 1.10
8	Common & Continuity	P5520 g	COMMON	From 0.96 to 1.10
9	Jumper	P5520 a	COMMON	From 0.96 to 1.10
10	Continuity	P5520 n	COMMON	From 0.96 to 1.10
11	Instrument Wiring	P5520 P	COMMON	From 0.96 to 1.10
12	Instrument Wiring	P5520 R	COMMON	From 0.96 to 1.10

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Table XIII (Cont)

LEAKAGE-2 Test

CIRCUIT SELECT Switch Position	Function	Afterbody Connector Pin		Resistance Value vs Lamp Indication
		From	To	NO GO lamp illuminates at 0.95 megohm and indication changes to GO lamp before 1.10 megohms is reached
13	Instrument Wiring	P5520 S	COMMON	From 0.96 to 1.10
14	Instrument Wiring	P5520 T	COMMON	From 0.96 to 1.10
15	Instrument Wiring	P5520 U	COMMON	From 0.96 to 1.10
16	Igniter	P5520 i	COMMON	From 0.96 to 1.10
17	Igniter	P5520 i	COMMON	From 0.96 to 1.10
18	Seawater Battery	P5520 W	COMMON	From 0.96 to 1.10
19	Spare	N/C	N/C	0-infinity
20	Spare	N/C	N/C	0-infinity
21	Spare	N/C	N/C	0-infinity
22	Spare	N/C	N/C	0-infinity
23	Spare	N/C	N/C	0-infinity
24	OFF	---	---	

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4.7.3.6 Actuator control test

4.7.3.6.1 Control settings. Position test set panel controls to the following settings:

Control	Position
CIRCUIT SELECT	OFF
TEST SELECT	ACTUATOR CONTROL
SELF TEST	OFF
SPEED	HI
DIRECTION	OFF REMOTE
SELECT	UPPER RUDDER
OVERLOAD ON	ON
POWER	ON

4.7.3.6.2 Perform actuator power control test by selecting control settings and observing indications specified in Table XIV.

NOTE: The DIRECTION switch initiates the test cycle. Position DIRECTION switch as specified after other controls have been positioned.

Table XIV

Actuator Power Control Tests

Control		Lamp Indication
DIRECTION	SELECT	
LEFT/UP	UPPER RUDDER	LEFT/UP, TEST IN PROGRESS, and NO GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and LEFT/UP remain illuminated.

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Table XIV (Cont)

Actuator Power Control Tests

Control		Lamp Indication
DIRECTION	SELECT	
RIGHT/DOWN	UPPER RUDDER	RIGHT/DOWN, TEST IN PROGRESS, and NO GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and RIGHT/DOWN remain illuminated.
OFF REMOTE	LOWER RUDDER	
LEFT/UP	LOWER RUDDER	LEFT/UP, TEST IN PROGRESS, and NO GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and LEFT/UP remain illuminated.
RIGHT/DOWN	LOWER RUDDER	RIGHT/DOWN, TEST IN PROGRESS, and NO GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and RIGHT/DOWN remain illuminated.
OFF REMOTE	ELEVATOR	
LEFT/UP	ELEVATOR	LEFT/UP, TEST IN PROGRESS, and NO GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and LEFT/UP remain illuminated.

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Table XIV (Cont)

Actuator Power Control Tests

Control		Lamp Indication
DIRECTION	SELECT	
RIGHT/DOWN	ELEVATOR	RIGHT/DOWN, TEST IN PROGRESS, and NO GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and RIGHT/DOWN remain illuminated.
OFF REMOTE	OFF	

4.7.3.6.3 Connect digital dc voltmeter plus input to P5520 pin C and return to pin B. Shunt pins with a 41.0 ohm resistor, plus or minus 0.25 percent, 20 watts or greater. Set oscilloscope controls as follows:

Control	Position
TRIGGER MODE	AC
TRIG SLOPE	INT -
TIME/CM	20 ms
VOLTS/CM	10
HORIZONTAL DISPLAY	(A) SINGLE SWEEP
MODE	A ONLY
RESET (pushbutton)	Depress

4.7.3.6.4 Connect oscilloscope Channel A to INHIBIT test point on test set front panel.

4.7.3.6.5 Within limits current test shall be performed by selecting control settings and observing indications specified in Table XV.

NOTE: Repeat each test as necessary by positioning DIRECTION switch to OFF-REMOTE and then returning DIRECTION to position specified.

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Table XV

Actuator Within Limits Currents Test

Control		Lamp Indication	Test Indication
DIRECTION	SELECT		
LEFT/UP	UPPER RUDDER	LEFT/UP, TEST IN PROGRESS, and GO illuminate GO and TEST IN PROGRESS extinguish after 2 to 4 seconds. NO GO and LEFT/UP illuminate.	Single trace on oscilloscope presents a negative pulse of 25 to 30 v amplitude and 60 to 100 ms in width. Digital dc voltmeter indicates +37.5 to +38.5 vdc during time period TEST IN PROGRESS lamp is illuminated. Pin C positive.
Disconnect 41 ohm resistor and dc voltmeter + input from pin C connector J5520 and connect them to pin A of same connector. Depress RESET pushbutton on oscilloscope.			
RIGHT/DOWN	UPPER RUDDER	RIGHT/DOWN, TEST IN PROGRESS and GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and RIGHT/DOWN remain illuminated.	Single trace on oscilloscope presents a negative pulse of 25 to 30 v amplitude and 60 to 100 ms in width. Digital dc voltmeter indicates -37.5 to -38.5 vdc during time period TEST IN PROGRESS lamp is illuminated.
OFF REMOTE	LOWER RUDDER		
Disconnect resistor. Connect 41-ohm resistor and digital dc voltmeter from pin J connector P5520 to pin K of same connector. Pin K is positive. Depress RESET on oscilloscope.			

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Table XV (Cont)

Actuator Within Limits Currents Test

Control		Lamp Indication	Test Indication
DIRECTION	SELECT		
LEFT/UP	LOWER RUDDER	LEFT/UP, TEST IN PROGRESS, and GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and LEFT/UP remain illuminated.	Single trace on oscilloscope presents a negative pulse of 25 to 30 v amplitude and 60 to 100 ms in width. Digital dc voltmeter indicates +37.5 to +38.5 vdc during time period TEST IN PROGRESS lamp is illuminated. Pin K positive.
Disconnect 41-ohm resistor and digital dc voltmeter lead from pin J, connector P5520, and connect them to pin L of same connector. Depress RESET on oscilloscope.			
RIGHT/DOWN	LOWER RUDDER	RIGHT/DOWN, TEST IN PROGRESS, and GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and RIGHT/DOWN remain illuminated.	Single trace on oscilloscope presents a negative pulse of 25 to 30 v in amplitude and 60 to 100 ms in width. Digital dc voltmeter indicates -37.5 to -38.5 vdc during period TEST IN PROGRESS lamp is illuminated.
OFF REMOTE	ELEVATOR		
Disconnect 41-ohm resistor. Connect a 152 ohm ±0.25% 10 watt resistor from pin F of connector J5520 to pin E of the same connector. Connect digital dc voltmeter in parallel with resistor. Pin F is positive. Depress RESET button on oscilloscope.			

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Table XV (Cont)

Actuator Within Limits Currents Test

Control		Lamp Indication	Test Indication
DIRECTION	SELECT		
LEFT/UP	ELEVATOR	LEFT/UP, TEST IN PROGRESS, and GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and LEFT/UP remain illuminated.	Single trace on oscilloscope presents a negative pulse of 25 to 30 v amplitude and 60 to 100 ms in width. Digital dc voltmeter indicates +37.5 to +38.5 vdc during period TEST IN PROGRESS lamp is illuminated. Pin F positive.
Disconnect 152 ohm resistor and digital voltmeter lead from pin E of connector J5520 and connect them to pin G of same connector. Depress RESET pushbutton on oscilloscope.			
RIGHT/DOWN	ELEVATOR	RIGHT/DOWN, TEST IN PROGRESS and GO illuminate. TEST IN PROGRESS extinguishes after 2 to 4 seconds. NO GO and RIGHT/DOWN remain illuminated.	Single trace on oscilloscope presents a negative pulse of 25 to 30 v in amplitude and 60 to 100 ms in width. Digital dc voltmeter indicates -37.5 to -38.5 vdc during period TEST IN PROGRESS lamp is illuminated.
OFF REMOTE	OFF		

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4.7.3.6.6 REMOTE CONTROL and LO SPEED test control settings. Position test set panel controls to the following settings:

Control	Position
CIRCUIT SELECT	OFF
TEST SELECT	ACTUATOR CONTROL
SELF TEST	OFF
SPEED	HI
DIRECTION	OFF REMOTE
SELECT	UPPER RUDDER
OVERLOAD ON	ON
POWER	ON

4.7.3.6.7 Connect digital dc voltmeter, shunted with a 41 ohm resistor, positive lead to P5520 pin C and negative lead to pin B.

4.7.3.6.8 Position SPEED switch to LO position; TEST IN PROGRESS lamp shall illuminate.

4.7.3.6.9 Perform REMOTE CONTROL and LO SPEED test by selecting control settings and observing indications specified in Table XVI. Depress the pushbutton switch indicated in the REMOTE CONTROL column.

Table XVI

REMOTE CONTROL and LO SPEED Test

Control		Test Indication
REMOTE CONTROL	SELECT	
LEFT/UP	UPPER RUDDER	Digital dc voltmeter indicates +11.5 to +14.5 vdc while switch is depressed.
Disconnect voltmeter and 41 ohm resistor leads from pin C and connect them to pin A.		
RIGHT/DOWN	UPPER RUDDER	Digital dc voltmeter indicates -11.5 to -14.5 vdc while switch is depressed.

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Table XVI (Cont)

REMOTE CONTROL and LO SPEED Test

Control		Test Indication
REMOTE CONTROL	SELECT	
Disconnect voltmeter leads and resistor. Connect voltmeter, shunted by 41 ohm resistor, positive lead to J5520 pin K and negative lead to pin J.		
LEFT/UP	LOWER RUDDER	Digital dc voltmeter indicates +11.5 to +14.5 vdc while switch is depressed.
Disconnect voltmeter and 41 ohm resistor leads from pin J and connect them to pin L.		
RIGHT/DOWN	LOWER RUDDER	Digital dc voltmeter indicates -11.5 to -14.5 vdc while switch is depressed.

4.7.3.6.10 Disconnect oscilloscope, digital dc voltmeter, and resistor.

4.7.3.6.11 Connect a 38.5 ohm ±0.25% 20 watt resistor between pin B and pin C of connector J5520.

4.7.3.6.12 Set SPEED switch to HI position.

4.7.3.6.13 Perform out of limits current test according to control position settings and indications specified in Table XVII as follows:

Table XVII

Actuator Out of Limits Current Test

Control		Lamp Indication
DIRECTION	SELECT	
OFF REMOTE	UPPER RUDDER	
LEFT/UP		LEFT/UP, TEST IN PROGRESS, and GO illuminate. GO extinguishes immediately and NO GO illuminates.

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Table XVII (Cont)

Actuator Out of Limits Current Test

Control		
DIRECTION	SELECT	Lamp Indication
LEFT/UP (Cont)		After a 2 to 4 second time lapse TEST IN PROGRESS extinguishes. NO GO and LEFT/UP remain illuminated.
Disconnect 38.5 ohm resistor from pin C and connect it to pin A of connector J5520.		
RIGHT/DOWN	UPPER RUDDER	RIGHT/DOWN, TEST IN PROGRESS, and GO illuminate. GO extinguishes immediately and NO GO illuminates. After a 2 to 4 second time lapse TEST IN PROGRESS extinguishes. NO GO and RIGHT/DOWN remain illuminated.
Disconnect resistor. Connect 38.5 ohm resistor from pin K to pin L connector P5520.		
OFF REMOTE	LOWER RUDDER	
RIGHT/DOWN		RIGHT/DOWN, TEST IN PROGRESS, and GO illuminate. GO extinguishes immediately and NO GO illuminates. After a 2 to 4 second time lapse TEST IN PROGRESS extinguishes. NO GO and RIGHT/DOWN remain illuminated.

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Table XVII (Cont)

Actuator Out of Limits Current Test

Control		Lamp Indication
DIRECTION	SELECT	
Disconnect 38.5 ohm resistor from pin L connector P5520 and connect it to pin J of same connector.		
LEFT/UP	LOWER RUDDER	LEFT/UP, TEST IN PROGRESS, and GO illuminate. GO extinguishes immediately and NO GO illuminates. After a 2 to 4 second time lapse TEST IN PROGRESS extinguishes. NO GO and LEFT/UP remain illuminated.
Disconnect 38.5 ohm resistor. Connect a 146 ohm ±0.25% at 10 watt resistor from pin F connector P5520 to pin E of same connector.		
OFF REMOTE	ELEVATOR	LEFT/UP, TEST IN PROGRESS, and GO illuminate. GO extinguishes immediately and NO GO illuminates. After a 2 to 4 second time lapse TEST IN PROGRESS extinguishes. NO GO and LEFT/UP remain illuminated.
Disconnect 146 ohm resistor from pin E connector J5520 and connect it to pin G of same connector.		
RIGHT/DOWN	ELEVATOR	RIGHT/DOWN, TEST IN PROGRESS, and GO illuminate. GO extinguishes immediately and NO GO illuminates.

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Table XVII (Cont)

Actuator Out of Limits Current Test

Control		Lamp Indication
DIRECTION	SELECT	
RIGHT/DOWN (Cont)		After a 2 to 4 second time lapse TEST IN PROGRESS extinguishes. NO GO and RIGHT/DOWN remain illuminated.
OFF REMOTE	OFF	

4.7.3.6.13 Disconnect resistor from connector pins.

4.7.3.7 Timing tests

4.7.3.7.1 Control settings. Position electronic counter controls as follows:

Control	Position
TRIGGER INPUT	SEP
TRIGGER SLOPE-START	+
TRIGGER LEVEL-START	0 VOLT LEVEL
TRIGGER SLOPE-STOP	-
TRIGGER LEVEL-STOP	+30 VOLT LEVEL
FUNCTION SELECTOR	TIME INTERVAL
TIME UNIT	MILLISEC x 100

Position test set front controls as follows:

Control	Position
CIRCUIT SELECT	OFF
TEST SELECT	ACTUATOR CONTROL
SELF TEST	OFF

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Control (Cont)	Position (Cont)
SPEED	HI
DIRECTION	OFF REMOTE
SELECT	OFF
OVERLOAD ON	ON
POWER	ON

4.7.3.7.2 Connect TRIGGER INPUT-START on electronic counter to ACTUATOR test point on front panel. Connect TRIGGER INPUT-STOP to TIMER test point on front panel. Connect the shields from the two leads to COMMON test point on the front panel.

4.7.3.7.3 Connect a 41 ± 0.25% ohm 20 watt resistor between pins B and C of connector J5520.

4.7.3.7.4 Perform timing accuracy test by selecting control settings and observing indications specified in Table XVIII:

Table XVIII

Timing Accuracy Test

Control		Counter Indication
DIRECTION	SELECT	
LEFT/UP	UPPER RUDDER	690 to 710 milliseconds
Disconnect 41 ohm resistor from pin C and connect it to pin A of connector J5520.		
RIGHT/DOWN	UPPER RUDDER	690 to 710 milliseconds
OFF REMOTE	UPPER RUDDER	- - -
Position STOP SLOPE control on counter to + SLOPE at +30 VOLT LEVEL.		
RIGHT/DOWN	UPPER RUDDER	1035 to 1065 milliseconds
Disconnect 41 ohm resistor from pin A and connect to pin C of J5520.		
LEFT/UP	UPPER RUDDER	1035 to 1065 milliseconds
OFF REMOTE		

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4.7.3.7.5 Timing indication test control positions of oscilloscope with CA preamplifier shall be set as follows:

Control	Position
TRIGGER SLOPE	EXT +
TRIGGERING MODE	AC
HORIZONTAL DISPLAY	'A' SINGLE SWEEP
TIME/CM	.20 sec.
VOLTS/CM (Chan A)	.5
VOLTS/CM (Chan B)	10
MODE	CHOPPED

4.7.3.7.6 Oscilloscope Channel A preamp lead shall be connected to EXT TRIGGER on oscilloscope and ACTUATOR test point on test set front panel. Channel B preamp lead shall be connected to TIMER test point on front panel.

4.7.3.7.7 Connect a 41 ohm ±0.25% 20 watt resistor between pins A and B of connector J5520.

4.7.3.7.8 Depress oscilloscope RESET pushbutton switch.

4.7.3.7.9 Position test set front panel DIRECTION switch to RIGHT/DOWN.

4.7.3.7.10 Observe negative rectangular pulse on Channel B, starting 700 milliseconds after sweep start with a width of 350 milliseconds.

4.7.3.7.11 Observe NO GO indication after TEST IN PROGRESS lamp extinguishes.

4.7.3.7.12 Position DIRECTION switch to OFF REMOTE, then to RIGHT/DOWN.

4.7.3.7.13 While trace is sweeping, position DIRECTION switch to OFF REMOTE.

4.7.3.7.14 Observe that Trace A returns to baseline as DIRECTION switch is returned to OFF REMOTE.

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4.7.3.7.15 Repeat steps 4.7.3.7.12 through 4.7.3.7.14 as necessary, observing GO-NO GO indications after oscilloscope sweep and TEST IN PROGRESS lamp extinguishes. GO lamp shall remain illuminated when DIRECTION switch is turned to OFF REMOTE as indicated on Channel A during negative pulse on Channel B. NO GO lamp shall illuminate when DIRECTION switch is turned to OFF REMOTE before and after negative pulse appears on Channel B.

4.7.3.7.16 Disconnect oscilloscope and resistor from test set.

4.7.3.8 Flash field test

4.7.3.8.1 Position POWER switch on test set front panel to ON.

4.7.3.8.2 Connect dc milliammeter from pin N to pin R of connector J5510 with positive (+) lead on pin R. Set range to 1000 milliamperes. Position POWER switch to ON.

4.7.3.8.3 Depress FLASH FIELD pushbutton switch.

4.7.3.8.4 Ammeter shall indicate 850 to 910 milliamperes.

4.7.3.8.5 Disconnect ammeter from connector J5510 pins.

4.7.3.9 ON-OVERLOAD switch operation

4.7.3.9.1 Control settings. Position controls on test set front panel as follows:

Control	Position
CIRCUIT SELECT	OFF
TEST SELECT	ACTUATOR CONTROL
SELF TEST	OFF
SPEED	HI
DIRECTION	RIGHT/DOWN
SELECT	UPPER/RUDDER
ON-OVERLOAD	ON
POWER	ON

CAUTION: Plus 40 vdc exists at pin B of connector J5520; prevent accidental shorting to ground.

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4.7.3.9.2 Connect a lead to pin B of connector J5520. Connect a second lead to COMMON test point on test set front panel.

4.7.3.9.3 Short the two leads of 4.7.3.9.2 together. The ON-OVERLOAD switch shall trip to overload position within three seconds.

4.7.3.9.4 Position DIRECTION switch to OFF REMOTE. Disconnect leads from test set and connector. Return ON-OVERLOAD switch to ON position.

4.7.3.9.5 Check all PRESS TO TEST lamps by depressing pushbuttons. Each, with the exception of the POWER indicator, shall illuminate when depressed. Position POWER switch to OFF.

4.7.3.10 Backlash gage shall perform in accordance with the requirements of Drawing 2541215.

4.7.3.11 Torque gage shall perform in accordance with the requirements of Drawing 2541265.

4.7.3.12 Remote control cable W2 shall perform in accordance with the requirements of Drawing 2541204.

4.7.3.13 Auxiliary cable W4 shall perform in accordance with the requirements of Drawing 2541206.

4.7.3.14 Extension board shall perform in accordance with the requirements of Drawing 2541198.

4.7.3.15 Disconnect all cables from test set front panel.

4.7.4 Examination of product. The test set shall be examined for conformance to the requirements specified herein and on the drawings listed in DL 2541200. The presence of any one of the following defects shall constitute a failure:

4.7.4.1 Test set not complete; component parts missing.

4.7.4.2 Component parts not new, evidence of reclaimed, used, or salvaged components.

4.7.4.3 Evidence of unauthorized materials or repairs.

4.7.4.4 Receptacles do not accept mating plugs.

4.7.4.5 Panel inscriptions incomplete, incorrect, or illegible.

4.7.4.6 Case not watertight.

4.7.4.7 Continuity not in accordance with schematic.

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4.7.4.8 Test set not correctly identified.

4.7.4.9 Test set not preserved and packaged in accordance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A. Preservation and packaging of the test set for Level A shipment shall be in accordance with MIL-P-116. Cleaning shall be accomplished in accordance with Process C-1 and drying, if required, shall be accomplished in accordance with Procedure D-4. Equipment case shall conform to Method IId of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging of the test set shall be in accordance with the supplier's own commercial practice, when such practices have been proven by testing to conform to all applicable requirements.

5.2 Packing. Packing shall be Level A or C, as specified (see 6.2).

5.2.1 Level A. The test set, preserved and packaged in accordance with 5.1, shall be overpacked in containers in compliance with PPP-B-636, Type 1, Class 2.

5.2.2 Level B. Not applicable.

5.2.3 Level C. The test set, when preserved and packaged as specified in 5.1, shall be overpacked to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Interstate Commerce Commision regulations and with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2.4 Cushioning or dunnage. The supplier shall provide sufficient cushioning or dunnage material to adequately protect equipment from damage due to shock or vibration loads encountered during handling and transportation, and to ensure safe delivery to the receiving activity.

5.3 Marking

5.3.1 Normal marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 Description and intended use

6.1.1 Description of test set. The test set is a portable unit of electronic test equipment contained in a watertight, shock protected combination case with a removable cover and facilities for storage of special tools and cabling. The case also provides protection from environmental deterioration while in storage and damage from handling during shipment. The test set front panel provides controls and connectors accessible for operation and interconnection of the test set when the cover is removed.

6.1.2 Intended use. The test set is to be used in the Fleet as electronic test equipment for checkout and test of a torpedo afterbody.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Required level of preservation and packaging.

(c) Required level of packing.

(d) Assigned activity for preproduction and periodic tests.

(e) Required data.

(f) Any exceptions to this specification.

(g) All additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions

6.3.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement given in one half the peak-to-peak excursion in inches or the corresponding maximum acceleration in gravity units (g). The frequency is given in cycles per second (cps).

6.3.2 Shock. Shock is an abrupt change in the position of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse force in units of gravitational acceleration (g's), of duration in units of time, milliseconds (ms), and a shock wave approximating a one-half sine wave. The time duration shall be measured between two points where two lines intersect the zero "g" line. These lines are tangent to the shock wave slopes.

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6.3.3 Temperature. The temperature of the environment to which the equipment may be exposed, expressed in degrees Fahrenheit (°F).

6.3.4 Humidity. The relative humidity of the environment to which the test set is exposed, expressed as a percent.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

(a) Level A	This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.
(b) Level B	Not applicable.
(c) Level C	This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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REPORT FORM

QUALITY ASSURANCE PROVISION

ACCEPTANCE TESTS AND EXAMINATIONS

FOR

AFTERBODY TEST SET

DL 2541200

(For Torpedo Mk 46 Mod 0/1)

EQUIPMENT STATUS			
CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

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REPORT FORM

QUALITY ASSURANCE PROVISION

ACCEPTANCE TEST AND EXAMINATION

AFTERBODY TEST SET DL 2541200

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the acceptance tests and examinations of 4.7 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the test set and bear the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. The acceptance tests and examinations of 4.7 shall be completed and the results recorded or verified in this report form prior to acceptance of the test set. Unless otherwise noted the sequence of testing, recording, and verification, as performed in 4.7, is mandatory and must be performed in the order shown.

20.2 Recording of results. Where space is provided to indicate the value of specific reading obtained, the specific reading shall be recorded if a limit or limits is given in the Limits column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limits column, the individual performing the test or inspection shall initial in the space provided in the Reading column indicating that the requirement has been met satisfactorily.

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30. ACCEPTANCE TEST AND EXAMINATION

	Reference	Limits	Reading
30.1 Test equipment	4.7.2	In Calibration	_____
30.2 Test procedure	4.7.3		
30.2.1 Equipment turn-on	4.7.3.2		
30.2.1.1 Test set power	4.7.3.2.4	Illuminates	_____
30.2.2 Self test	4.7.3.3		
30.2.2.1 +40 volt test	4.7.3.3.2	+39.6 to +40.4 vdc	_____
GO 1		GO Illuminates	_____
30.2.2.2 GO 2	4.7.3.3.5	GO Illuminates	_____
30.2.2.3 GO 3	4.7.3.3.7	GO Illuminates	_____
30.2.2.4 NO GO 4	4.7.3.3.9	NO GO Illuminates	_____
30.2.2.5 NO GO 5	4.7.3.3.11	NO GO Illuminates	_____
30.2.2.6 NO GO 6	4.7.3.3.15	NO GO, TEST IN PROGRESS, and LEFT/UP lamps illuminate	_____
		After 2 to 4 seconds TEST IN PROGRESS lamp shall extinguish while LEFT/UP and NO GO lamps remain illuminated.	_____
DC volts		+0.01 to -0.01 vdc	_____
30.2.2.7 NO GO 7	4.7.3.3.17	GO lamp flashes momentarily	_____
		NO GO, TEST IN PROGRESS, and RIGHT/DOWN lamps illuminate.	_____

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		Reference	Limits	Reading
30.2.2.7	(Continued)		After 2 to 4 seconds TEST IN PROGRESS lamp shall extinguish while RIGHT/DOWN and NO GO lamps remain illuminated.	_____ _____
	DC volts		+1.3 to +1.5 vdc	_____
30.2.2.8	SEE INST 8 test	4.7.3.3.19	GO, TEST IN PROGRESS and LEFT/UP lamps illuminate.	_____
			After 2 to 4 seconds TEST IN PROGRESS and GO lamps extinguish while NO GO lamp illuminates and LEFT/UP lamp remains illuminated.	_____ _____
	DC volts		800 to 1000 mv	_____
30.2.3	Continuity tests	4.7.3.4		
30.2.3.1	Continuity-1 tests	4.7.3.4.2		
	1. NO GO below		6.3 ohms	_____
	Change to GO		6.3-7.7 ohms	_____
	GO		7.7 to 38 ohms	_____
	Change to NO GO		38 to 42 ohms	_____
	2. NO GO below		6.3 ohms	_____
	Change to GO		6.3 to 7.7 ohms	_____
	GO		7.7 to 38 ohms	_____
	Change to NO GO		38 to 42 ohms	_____
	3. NO GO below		6.3 ohms	_____
	Change to GO		6.3 to 7.7 ohms	_____
	GO		7.7 to 66.5 ohms	_____
	Change to NO GO		66.5 to 73.5 ohms	_____

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	Reference	Limits	Reading
30.2.3.1 (Continued)			
4. NO GO below		18.8 ohms	_____
Change to GO		18.8 to 21.2 ohms	_____
GO		21.2 to 43.0 ohms	_____
Change to NO GO		43.0 to 47.0 ohms	_____
5. NO GO below		18.8 ohms	_____
Change to GO		18.8 to 21.2 ohms	_____
GO		21.2 to 43.0 ohms	_____
Change to NO GO		43.0 to 47.0 ohms	_____
6. NO GO below		9.0 ohms	_____
Change to GO		9.0 to 11.0 ohms	_____
GO		11.0 to 18.5 ohms	_____
Change to NO GO		18.5 to 21.5 ohms	_____
7. NO GO below		8.0 ohms	_____
Change to GO		8.0 to 9.0 ohms	_____
GO		9.0 to 11.0 ohms	_____
Change to NO GO		11.0 to 12.0 ohms	_____
8. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
9. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
10. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.1 to 5.0 ohms	_____
11. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
12. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
13. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
14. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
15. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____

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	Reference	Limits	Reading
30.2.3.1 (Continued)			
16.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
17.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
18.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
19.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
20.	GO	0 to infinity	_____
21.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
22.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
23.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
30.2.3.2 Continuity-2 tests	4.7.3.4.4		
1.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
2.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
3.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
4.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
5.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
6.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____
7.	GO Change to NO GO	0.0 to 1.0 ohm 1.0 to 5.0 ohms	_____ _____

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	Reference	Limits	Reading
30.2.3.2 (Continued)			
8. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
9. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
10. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
11. GO		0 to infinity	_____
12. GO		0 to infinity	_____
13. GO		0 to infinity	_____
14. GO		0 to infinity	_____
15. GO		0 to infinity	_____
16. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
17. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
18. GO		0.0 to 1.0 ohm	_____
Change to NO GO		1.0 to 5.0 ohms	_____
19. GO		0 to infinity	_____
20. GO		0 to infinity	_____
21. GO		0 to infinity	_____
22. GO		0 to infinity	_____
23. GO		0 to infinity	_____
30.2.4 Leakage tests	4.7.3.5		
30.2.4.1 Leakage-1 tests	4.7.3.5.2		
1. NO GO		0.95 megohms	_____
Change to GO		0.96 to 1.10 megohms	_____

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	Reference	Limits	Reading
30.2.4.1 (Continued)			
2. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
3. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
4. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
5. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
6. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
7. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
8. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
9. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
10. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
11. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
12. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
13. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
14. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
15. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____
16. NO GO Change to GO		0.95 megohms 0.96 to 1.10 megohms	_____ _____

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		Reference	Limits	Reading
30.2.4.1	(Continued)			
	17. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	18. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	19. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	20. GO		0 to infinity	_____
	21. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	22. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	23. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
30.2.4.2	Leakage-2 tests	4.7.3.5.3		
	1. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	2. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	3. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	4. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	5. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	6. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____
	7. NO GO		0.95 megohms	_____
	Change to GO		0.96 to 1.10 megohms	_____

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	Reference	Limits	Reading
30.2.4.2 (Continued)			
8.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
9.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
10.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
11.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
12.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
13.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
14.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
15.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
16.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
17.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
18.	NO GO Change to GO	0.95 megohms 0.96 to 1.10 megohms	_____ _____
19.	GO	0 to infinity	_____
20.	GO	0 to infinity	_____
21.	GO	0 to infinity	_____
22.	GO	0 to infinity	_____
23.	GO	0 to infinity	_____
24.	OFF		

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	Reference	Limits	Reading
30.2.5 Actuator control test	4.7.3.6		
30.2.5.1 Actuator power control tests	4.7.3.6.2		
(a) LEFT/UP, UPPER RUDDER switch pos.		LEFT/UP, TEST IN PROGRESS, and NO GO lamps illuminate	_____
(b)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(c)		NO GO, LEFT/UP lamps remain illuminated	_____
(d) RIGHT/DOWN UPPER RUDDER switch pos.		RIGHT/DOWN, TEST IN PROGRESS and NO GO lamps illuminate	_____
(e)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(f)		NO GO, RIGHT/DOWN lamps remain illuminated	_____
(g) LEFT/UP, LOWER RUDDER switch pos.		LEFT/UP, TEST IN PROGRESS, and NO GO lamps illuminate	_____
(h)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(i)		NO GO, LEFT/UP lamps remain illuminated	_____
(j) RIGHT/DOWN, LOWER RUDDER switch pos.		RIGHT/DOWN, TEST IN PROGRESS, and NO GO lamps illuminate	_____

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	Reference	Limits	Reading
30.2.5.1 (Continued)			
(k)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(l)		NO GO, RIGHT/DOWN lamps remain illuminated	_____
(m) LEFT/UP, ELEVATOR switch pos.		LEFT/UP, TEST IN PROGRESS, and NO GO lamps illuminate	_____
(n)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(p)		NO GO, LEFT/UP lamps remain illuminated	_____
(q) RIGHT/DOWN ELEVATOR switch pos.		RIGHT/DOWN, TEST IN PROGRESS and NO GO lamps illuminate	_____
(r)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(s)		NO GO, RIGHT/DOWN lamps remain illuminated	_____
30.2.5.2 Actuator within limits current test	4.7.3.6.5		
(a) LEFT/UP, UPPER RUDDER switches		LEFT/UP, TEST IN PROGRESS and GO lamps illuminate	_____
(b)		Single trace on oscilloscope presents a negative pulse of 25 to 30 volts amplitude and 60 to 100 ms in width	_____

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	Reference	Limits	Reading
30.2.5.2 (Continued)			
	(c)	Digital dc voltmeter indicates +37.5 to +38.5 vdc during time period TEST IN PROGRESS lamp is illuminated	_____
		Pin C Positive	_____
	(d)	GO lamp and TEST IN PROGRESS lamp extinguish after 2 to 4 seconds	_____
	(e)	NO GO lamp illuminates, LEFT/UP lamp remains illuminated	_____
	(f) RIGHT/DOWN UPPER RUDDER switches	RIGHT/DOWN, TEST IN PROGRESS and GO lamps illuminate	_____
	(g)	Single trace on oscilloscope presents a negative pulse of 25 to 30 volts amplitude and 60 to 100 ms in width	_____
	(h)	Digital dc voltmeter indicates -37.5 to -38.5 vdc during time period TEST IN PROGRESS lamp is illuminated	_____
	(i)	NO GO lamp illuminates, RIGHT/DOWN lamp remains illuminated	_____

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	Reference	Limits	Reading
30.2.5.2 (Continued)			
	(j)	NO GO lamp illuminates, RIGHT/DOWN lamp remains illuminated	_____
	(k) LEFT/UP, LOWER RUDDER switch pos.	LEFT/UP, TEST IN PROGRESS and GO lamps illuminate	_____
	(l)	Single trace on oscilloscope presents a negative pulse of 25 to 30 volts amplitude and 60 to 100 ms in width	_____
	(m)	Digital dc voltmeter indicates +37.5 to +38.5 vdc during time period TEST IN PROGRESS lamp is illuminated	_____
		Pin K Positive	_____
	(n)	GO lamp and TEST IN PROGRESS lamp extinguish after 2 to 4 seconds	_____
	(p)	NO GO lamp illuminates, LEFT/UP lamp remains illuminated	_____
	(q) RIGHT/DOWN LOWER RUDDER switch pos.	RIGHT/DOWN, TEST IN PROGRESS, and GO lamps illuminate	_____
	(r)	Single trace on oscilloscope presents a negative pulse of 25 to 30 volts amplitude and 60 to 100 ms in width	_____

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		Reference	Limits	Reading
30.2.5.2	(Continued)			
	(s)		Digital dc voltmeter indicates -37.5 to -38.5 vdc during time period TEST IN PROGRESS lamp is illuminated	_____
	(t)		GO lamp and TEST IN PROGRESS lamp extinguish after 2 to 4 seconds	_____
	(u)		NO GO lamp illuminates, RIGHT/DOWN lamp remains illuminated	_____
	(v) LEFT/UP, ELEVATOR switch pos.		LEFT/UP, TEST IN PROGRESS, and GO lamps illuminate	_____
	(w)		Single trace on oscilloscope presents a negative pulse of 25 to 30 volts amplitude and 60 to 100 ms in width	_____
	(x)		Digital dc voltmeter indicates +37.5 to +38.5 vdc during time period TEST IN PROGRESS lamp is illuminated	_____
			Pin F Positive	_____
	(y)		GO lamp and TEST IN PROGRESS lamp extinguish after 2 to 4 seconds	_____

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	Reference	Limits	Reading
30.2.5.2 (Continued)			
(z)		NO GO lamp illuminates, LEFT/UP lamp remains illuminated	_____
(aa) RIGHT/DOWN ELEVATOR switch pos.		RIGHT/DOWN, TEST IN PROGRESS, and GO lamps illuminate	_____
(ab)		Single trace on oscilloscope presents a negative pulse of 25 to 30 volts amplitude and 60 to 100 ms in width	_____
(ac)		Digital dc voltmeter indicates -37.5 to -38.5 vdc during time period TEST IN PROGRESS lamp is illuminated	_____
(ad)		GO lamp and TEST IN PROGRESS lamp extinguish after 2 to 4 seconds	_____
(ae)		NO GO lamp illuminates. RIGHT/DOWN lamp remains illuminated	_____
30.2.5.3 Remote control and LO speed test	4.7.3.6.9		
(a) TEST IN PROGRESS lamp illuminates			_____
(b) UPPER RUDDER LEFT/UP position		Voltmeter indicates +11.5 to +14.5 vdc	_____

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	Reference	Limits	Reading
30.2.5.3 (Continued)			
(c) UPPER RUDDER, RIGHT/DOWN switch positions		Voltmeter indicates -11.5 to -14.5 vdc	_____
(d) LOWER RUDDER, LEFT/UP switch positions		Voltmeter indicates +11.5 to +14.5 vdc	_____
(e) LOWER RUDDER, RIGHT/DOWN switch positions		Voltmeter indicates -11.5 to -14.5 vdc	_____
30.2.5.4 Actuator out of limits current test	4.7.3.6.12		
(a) LEFT/UP UPPER RUDDER switch position		LEFT/UP, TEST IN PROGRESS, and GO lamps illuminate	_____
(b)		GO lamp extinguishes immediately	_____
(c)		NO GO lamp illuminates	_____
(d)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(e)		NO GO, LEFT/UP lamps remain illuminated	_____
(f) RIGHT/DOWN UPPER RUDDER switch pos.		RIGHT/DOWN, TEST IN PROGRESS, and GO lamps illuminate	_____
(g)		GO lamp extinguishes immediately	_____
(h)		NO GO lamp illuminates	_____

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	Reference	Limits	Reading
30.2.5.4 (Continued)			
(i)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(j)		NO GO, RIGHT/DOWN lamps remain illuminated	_____
(k)	RIGHT/DOWN, LOWER RUDDER switch pos.	RIGHT/DOWN, TEST IN PROGRESS, and GO lamps illuminate	_____
(l)		GO lamp extinguishes immediately	_____
(m)		NO GO lamp illuminates	_____
(n)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(o)		NO GO, RIGHT/DOWN lamps remain illuminated	_____
(p)	LEFT/UP, LOWER RUDDER switch pos.	LEFT/UP, TEST IN PROGRESS, and GO lamps illuminate	_____
(q)		GO lamp extinguishes immediately	_____
(r)		NO GO lamp illuminates	_____
(s)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(t)		NO GO, LEFT/UP lamps remain illuminated	_____

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	Reference	Limits	Reading
30.2.5.4 (Continued)			
(u) LEFT/UP, ELEVATOR switch pos.		LEFT/UP, TEST IN PROGRESS, and GO lamps illuminate	_____
(v)		GO lamp extinguishes immediately	_____
(w)		NO GO lamp illuminates	_____
(x)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(y)		NO GO, LEFT/UP lamps remain illuminated	_____
(z) RIGHT/DOWN, ELEVATOR switch pos.		RIGHT/DOWN, TEST IN PROGRESS and GO lamps illuminate	_____
(aa)		GO lamp extinguishes immediately	_____
(ab)		NO GO lamp illuminates	_____
(ac)		TEST IN PROGRESS lamp extinguishes after 2 to 4 sec.	_____
(ad)		NO GO, RIGHT/DOWN lamps remain illuminated	_____
30.2.6 Timing tests	4.7.3.7		
30.2.6.1 Timing accuracy test	4.7.3.7.4		
(a) LEFT/UP, UPPER RUDDER switch pos.		Counter indicates 690 to 710 ms	_____

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	Reference	Limits	Reading
30.2.6.1 (Continued)			
(b) RIGHT/DOWN UPPER RUDDER switch pos.		Counter indicates 690 to 710 ms	_____
(c) RIGHT/DOWN, UPPER RUDDER switch pos.		Counter indicates 1035 to 1065 ms	_____
(d) LEFT/UP, UPPER RUDDER switch pos.		Counter indicates 1035 to 1065 ms	_____
30.2.6.3 Timing indication	4.7.3.7.5		
30.2.6.3.1 RIGHT/DOWN, UPPER RUDDER switches	4.7.3.7.9		
(a) DIRECTION switch OFF REMOTE (interruption during negative pulse)	4.7.3.7.15	GO lamp illuminates	_____
(b) DIRECTION switch OFF REMOTE (interruption before negative pulse)		NO GO lamp illuminates	_____
(c) DIRECTION switch OFF REMOTE (interruption after negative pulse)		NO GO lamp illuminates	_____
30.2.7 Flash, field test			
30.2.7.1 Depress FLASH FIELD pushbutton	4.7.3.8.4	DC milliameter shall indicate 850 to 910 ma	_____
		Pin R Positive	_____

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	Reference	Limits	Reading
30.2.8 ON-OVERLOAD switch operation	4.7.3.9		
30.2.8.1 Actuator section overload	4.7.3.9.3	ON-OVERLOAD switch shall trip to OVER-LOAD position within 3 sec.	_____
	4.7.3.9.4	GO or NO GO lamp shall illuminate	_____
30.2.8.2 PRESS TO TEST lamps	4.7.3.9.5	Illuminate when depressed	_____
30.2.9 Backlash gage	4.7.3.10	Performs according to drawing	_____
30.2.10 Torque gage	4.7.3.11	Performs according to drawing	_____
30.2.11 Remote control cable W2	4.7.3.12	Performs according to drawing	_____
30.2.12 Auxiliary cable W4	4.7.3.13	Performs according to drawing	_____
30.2.13 Extension board	4.7.3.14	Performs according to drawing	_____
30.3 Visual inspections	4.7.4		_____
30.3.1 Completeness	4.7.4.1		_____
30.3.2 Component parts	4.7.4.2		_____
30.3.3 Material and repairs	4.7.4.3		_____
30.3.4 Interchangeability	4.7.4.4		_____
30.3.5 Nomenclature, control and identification callouts	4.7.4.5		_____
30.3.6 Watertightness	4.7.4.6		_____
30.3.7 Wiring	4.7.4.7		_____
30.3.8 Identification	4.7.4.8		_____
30.3.9 Preservation and packaging	4.7.4.9		_____

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